

Purnendu Tiwari

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Education

Vellore Institute of Technology (VIT)
B.Tech in Computer Science; CGPA: 8.91/10.0

Bhopal, India
2023 – 2027

Relevant Coursework: Data Structures & Algorithms, OS, DBMS, OOP, Computer Networks.

Kendriya Vidyalaya No. 1
Class XII (CBSE); Percentage: 92.4%

Chhindwara, India
2021 – 2022

Technical Skills

Languages: Java, JavaScript, TypeScript

Web Development: React JS, Node JS, Express JS, MongoDB, MySQL, Next.js

Cloud Services: AWS (S3, CloudFront, IAM, VPC), Cloudflare

Developer Tools: Git, GitHub, Bitbucket, Jira, Docker, Postman

Methodologies: Agile, Scrum, SDLC

Experience

President

September 2024 – Present

Software Development Club, VIT Bhopal

Bhopal, India

- Spearheaded a cross-functional team of **6** members across Tech, Logistics, and Content divisions to execute club operations and long-term strategy.
- Orchestrated **5+** major campus events and hackathons, achieving a cumulative participation of over **300+** students.
- Organized a flagship seminar featuring a Google Manager, facilitating industry mentorship for **180+** attendees.
- Collaborated with the technical team to deploy the official SDC website, streamlining event registrations by **30%**.
- Recognized as a **Google CrowdSource Influencer** for driving community engagement.

Projects

sdcHouse – Real-time Virtual Space | *Next.js, WebSockets, WebRTC, Redis*

- Engineered a **metaverse-lite platform** featuring real-time spatial audio and video communication for 50+ concurrent users.
- Implemented **Socket.io** for state synchronization and **WebRTC** for peer-to-peer streaming, achieving sub-200ms latency.
- Optimized backend performance using **Redis** for pub/sub mechanisms, reducing server overhead during peak traffic by 25%.
- Developed a custom **spatial positioning algorithm** to manage media tracks, improving client-side CPU efficiency by 30%.

MediSin Care Platform | *React JS, Node JS, Python, AWS*

- Engineered a secure full-stack healthcare platform to manage sensitive medical data, implementing end-to-end encryption compliant with HIPAA/GDPR standards.
- Integrated Machine Learning models to analyze patient data and generate personalized healthcare recommendations.
- Increased user engagement by **40%** through the delivery of actionable health insights and a responsive UI.

Certifications

Introduction To Machine Learning – NPTEL (IIT Madras)

The Bits and Bytes of Computer Networking – Google

Achievements & Co-curricular

GirlScript Summer of Code: Ranked **419th** among global open-source contributors.

Youth Parliament: 1st Runner Up, Zonal-Level Competition.

National Debate: 2nd Runner Up, Ek Bharat Shresth Bharat Competition.

Music: Diploma in Classical Music (Raja Mansingh Tomar Music Arts University).